CLINICAL · EAR

Hearing tests: how to do and read them

Each hearing test answers a different question. Tuning forks give the type of loss at the bedside in seconds; the audiogram measures how much; the tympanogram shows what the middle ear is doing; and objective tests such as otoacoustic emissions and the brainstem response work when the patient cannot or will not respond.

AT A GLANCE

- Forks give type; audiogram gives amount; tympanogram shows the middle ear.
- Mask when ears differ.
- OAE and ABR suit those who can't respond.
- Reflexes help locate cochlear, VIII, VII and brainstem lesions.
- Read every audiogram in order: air, bone, compare ears, masking, fit the clinical picture.

1 CHAPTER ONE Overview

Which test answers which question.

1.1 The toolkit

HEARING TESTS

- BEDSIDE **Voice and tuning forks** conductive or sensorineural, roughly how much
- BEHAVIOURAL **Pure-tone and speech audiometry** thresholds and understanding
- MIDDLE EAR **Tympanometry and stapedial reflexes** drum, ossicles, tube, reflex arc
- OBJECTIVE **OAE, ABR, ASSR, ECochG** cochlea and neural pathway without a response

1.2 Choosing a test

Forks at the bedside; audiogram and tympanogram in clinic; OAE and ABR in newborns and uncooperative patients; ASSR for profound loss, beyond the range of

ABR.

CHAPTER TWO

2 Sound and the decibel

The language of audiology.

2.1 Decibel scales

- **dB SPL:** sound pressure relative to 20 µPa.
- **dB HL:** relative to average normal hearing at each frequency, so 0 dB HL is normal on the audiogram.
- **dB SL:** relative to the patient's own threshold.

2.2 Everyday levels

Whisper about 30 dB, conversation about 60 dB, a shout about 90 dB, discomfort about 120 dB and pain about 130 dB.

2.3 How the middle ear amplifies

The area ratio of the drum to the footplate (about 17 to 1) and the lever action of the ossicles (about 1.3 to 1) give a gain of about 22 times, around 25 dB. Without it, most sound energy would bounce off the fluid of the inner ear.

2.4 Where frequencies are heard

The base of the cochlea hears high frequencies, the apex low frequencies. The canal resonates at about 3000 Hz.

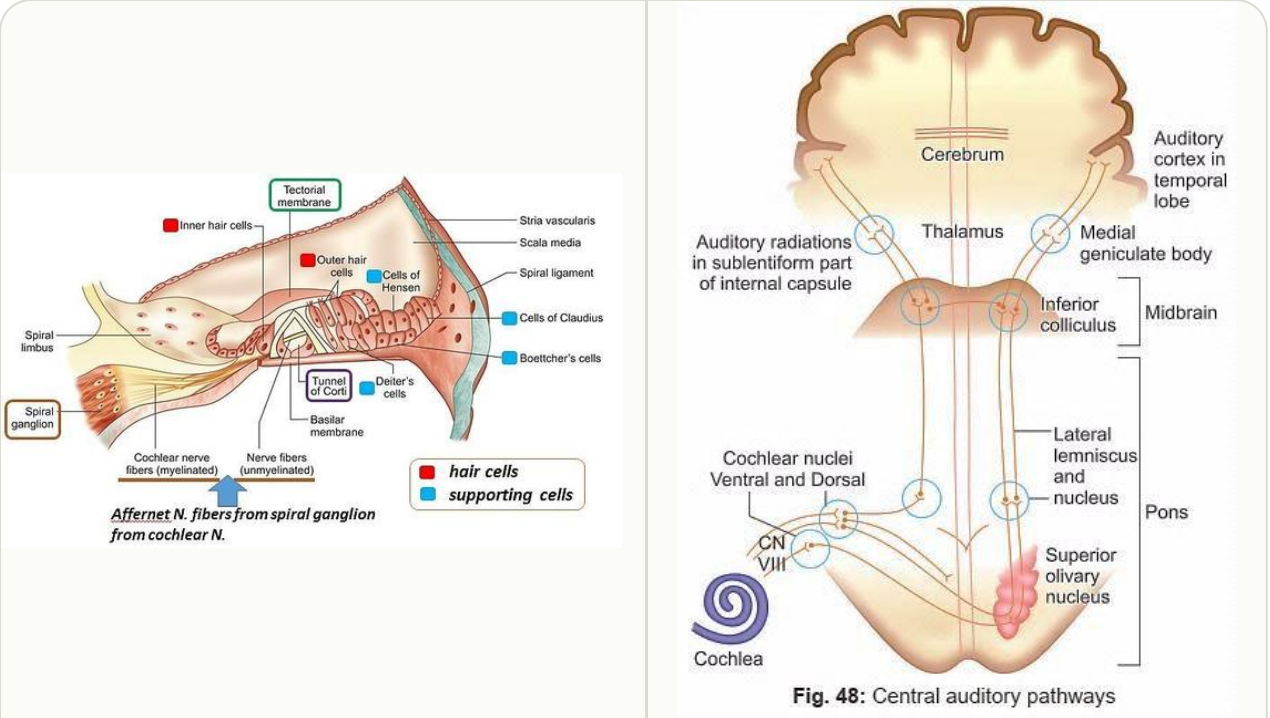


FIGURE 1 The organ of Corti, and the central auditory pathway that the ABR waves trace.
Anatomy atlas · ENT Steps p. 77, ENT Steps p. 84

CHAPTER THREE

3 Bedside tests

Voice and tuning forks.

3.1 Voice tests

A whisper heard at 6 m is taken as normal; the other ear is masked by rubbing the tragus.

3.2 Tuning fork tests

The 512 Hz fork is the routine choice: lower forks are felt as vibration, higher ones fade too quickly.

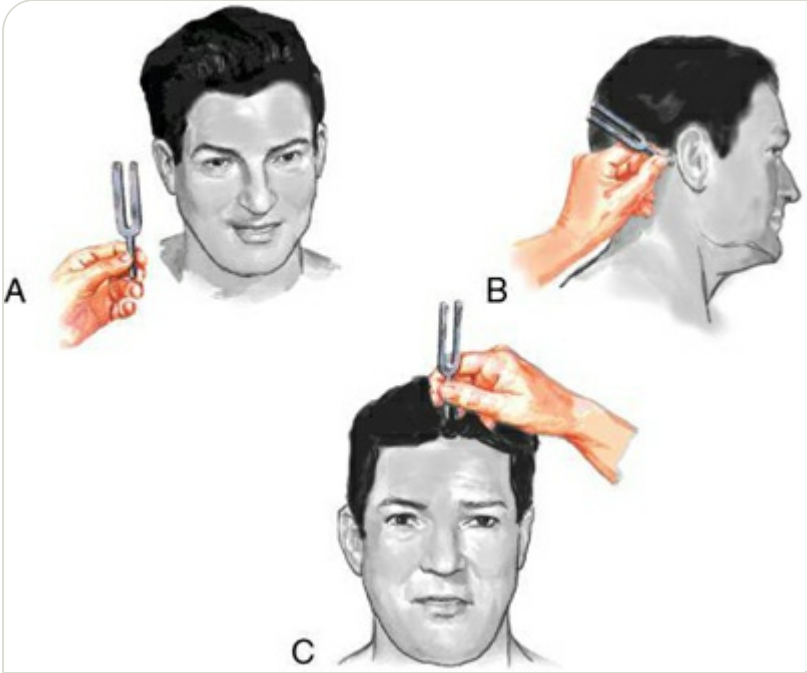


FIGURE 2 Tuning fork tests: (A) air conduction, (B) bone conduction, (C) the Weber test. Dhingra, fig. 4.1

3.2.1 Rinne test

Compares air conduction with bone conduction in the same ear. *Positive* (air louder) is normal, and also seen in sensorineural loss; *negative* (bone louder) means a conductive loss of at least about 20 dB.

3.2.2 Weber test

The fork on the forehead or vertex lateralises to the poorer ear in a conductive loss and to the better ear in a sensorineural loss. It is sensitive, lateralising with quite small differences between the ears.

3.2.3 Absolute bone conduction and Schwabach tests

Compare the patient's bone conduction with the examiner's (with the canal occluded in the ABC test). Reduced in sensorineural loss.

3.2.4 Bing and Gellé tests

Bing: occluding the canal makes the tone louder in normal and sensorineural ears but not in a conductive loss. Gellé: raising canal pressure normally reduces hearing, but not with a fixed or broken chain; tympanometry has replaced it.

READING TUNING FORKS

TEST	NORMAL	CONDUCTIVE	SENSORINEURAL
Rinne	Air louder	Bone louder	Air louder

TEST	NORMAL	CONDUCTIVE	SENSORINEURAL
Weber	Central	To poorer ear	To better ear
ABC	Same as examiner	Same	Reduced
Schwabach	Equal	Lengthened	Shortened

3.2.5 Estimating the gap from Rinne

A negative Rinne at 256 Hz only suggests a gap of about 20 to 30 dB; at 256 and 512 Hz, 30 to 45 dB; at all three forks, 45 to 60 dB.

WHERE THE BOOKS DIFFER · GAP THAT TURNS EACH FORK NEGATIVE

Dhingra
15 dB at 256, 30 dB at 512 and 45 dB at 1024 Hz.

Lee
15, 25 and 35 dB.

FALSE-NEGATIVE RINNE

In a severe one-sided sensorineural loss, the fork on the mastoid is heard by the other cochlea, so bone seems louder. Mask the good ear with a Barany noise box, and use the Weber.

4 CHAPTER FOUR Pure-tone audiometry

Thresholds, frequency by frequency.

4.1 How it is done

In a sound-treated room, the softest tone the patient hears at each frequency from 250 to 8000 Hz is recorded, first by air conduction through headphones, then by bone conduction with a vibrator on the mastoid.



FIGURE 3 A two-room audiometry set-up: the audiologist watches the patient through a glass partition.

Dhingra, fig. 4.2

4.2 The symbols

Frequency runs across the top from low (250 Hz) to high (8000 Hz), and loudness runs down the side, so a worse hearing threshold sits *lower* on the chart. By convention the right ear is red and the left ear blue. Air conduction is a circle (right) or a cross (left), joined by a line; bone conduction is an angle bracket pointing away from the ear tested, < for right and > for left, and a square bracket, [or], when the other ear was masked. An arrow below a symbol means no response at the machine's maximum output, which is about 120 dB for air but only about 60 to 70 dB for bone.

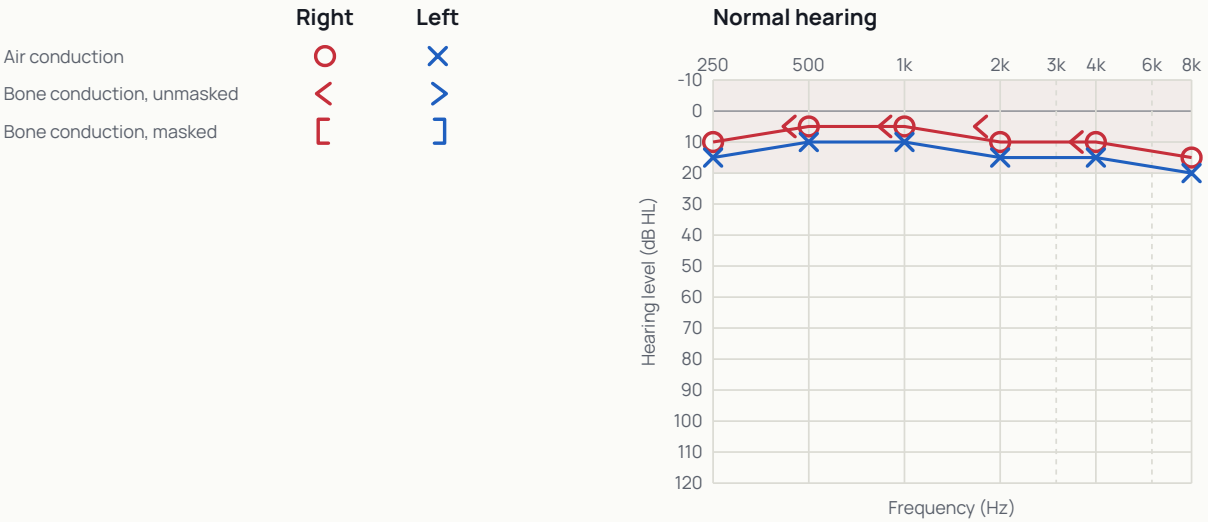


FIGURE 4 The symbols, and a normal audiogram: every threshold sits in the shaded band of 20 dB HL or better, and air and bone conduction coincide.

Drawn for ENT Atlas

4.3 Reading it, step by step

Read every audiogram in the same order, so nothing is missed.

READING A PURE-TONE AUDIOGRAM

1 Check the chart

→ right and left, masked or unmasked, reliability noted by the tester

2 Air conduction

→ normal (20 dB or better)

→ **how much loss**
mild, moderate, severe, profound

→ **which frequencies**
flat, sloping, rising, notched, cookie-bite

3 Bone conduction

→ **equal to air**
sensorineural

→ **better than air by more than 10 dB**

conductive component

→ **both raised with a gap**
mixed

4 **Compare the ears**

→ symmetrical

→ asymmetric by 15 dB or more at two adjacent frequencies, or 20 dB at one: MRI of the internal auditory meatus

5 **Check masking**

→ **ears differ by 40 dB or more on air, or any air–bone gap**
masked thresholds needed — was it done?

6 **Fit the clinical picture**

→ otoscopy, tuning forks and tympanometry must agree

→ if they do not, repeat or think non-organic

Bone conduction is cochlear function. The air–bone gap is the size of the conductive loss: a gap of up to about 60 dB is possible from the middle ear alone (complete ossicular discontinuity behind an intact drum), so a larger gap means the bone thresholds are wrong or the loss is mixed. Bone conduction can be *better than normal* in a third-window lesion such as superior canal dehiscence, giving an apparent conductive loss with normal tympanometry and present reflexes.

4.4 Masking

Sound presented to one ear can cross the skull and be heard by the other, so the non-test ear is masked with narrow-band noise. Interaural attenuation is 40 to 65 dB for air conduction with headphones (more with insert earphones) and close to 0 dB for bone, because the vibrator shakes the whole skull. Mask whenever the ears differ by 40 dB or more on air conduction, and for bone conduction whenever there is an air–bone gap in the test ear.

The usual method is the **plateau (Hood) technique**: noise in the non-test ear is raised in 10 dB steps while the threshold in the test ear is re-measured; while the threshold keeps rising with the noise, the other ear was answering, and once it stays the same across several steps (the plateau), the true threshold has been found. Without masking, a dead ear produces a **shadow curve**, a threshold that mirrors the good ear about 40 to 60 dB lower for air and almost level with it for bone. The **masking dilemma** arises in large bilateral conductive losses, as in bilateral atresia, when the noise needed to mask one ear crosses over and masks the test ear too; insert earphones and ABR by bone conduction help.

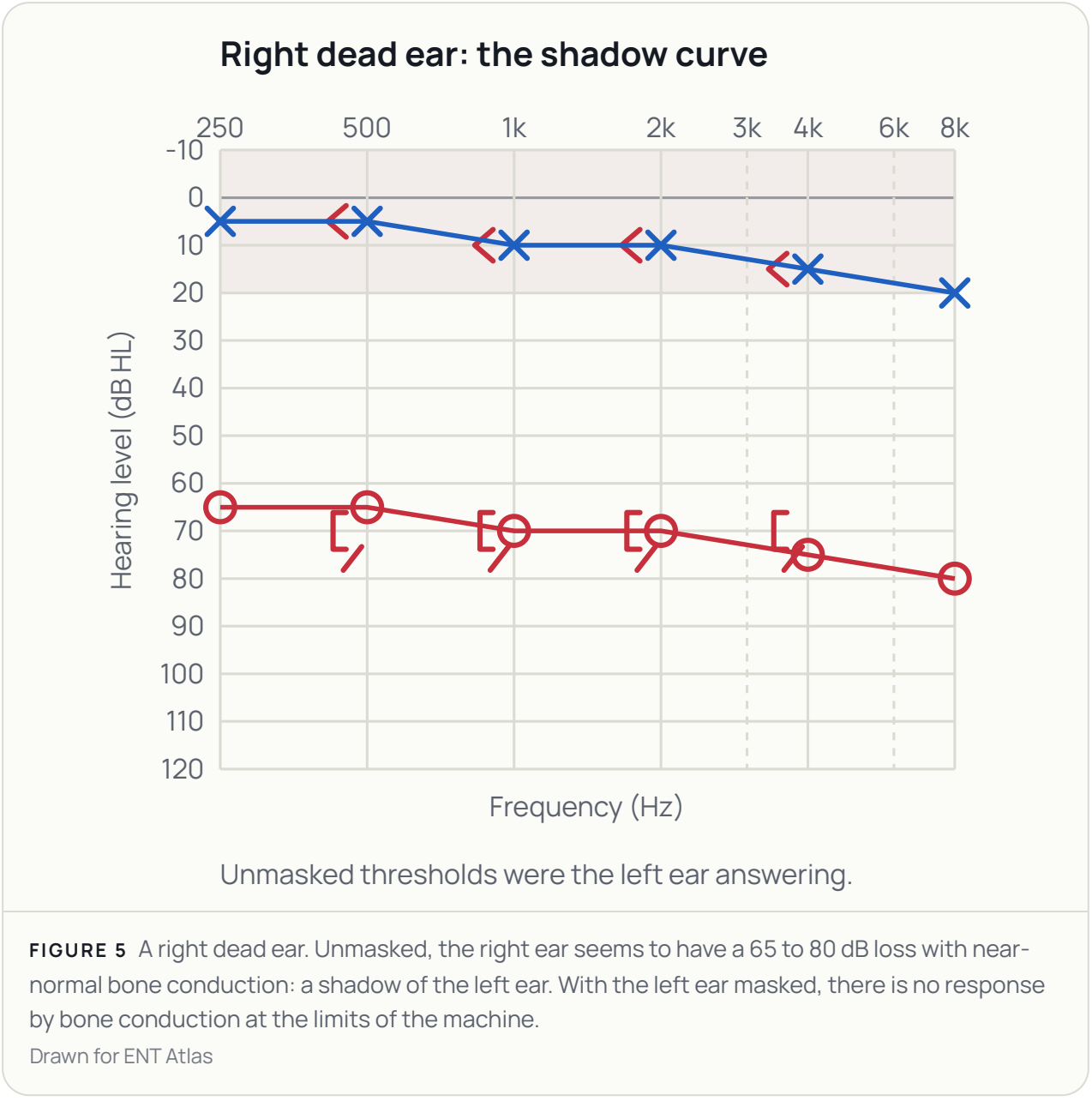


FIGURE 5 A right dead ear. Unmasked, the right ear seems to have a 65 to 80 dB loss with near-normal bone conduction: a shadow of the left ear. With the left ear masked, there is no response by bone conduction at the limits of the machine.

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4.5 Repeatability

Normal repeat audiograms agree within about 5 dB, so a change of 10 dB or more at two frequencies is significant. Grades of loss use the average of 500, 1000, 2000 and 4000 Hz in the better ear (see the hearing loss article).

5 CHAPTER FIVE Patterns on the audiogram

Worked examples.

5.1 Conductive and mixed losses

Otosclerosis produces a conductive loss, often worse in the low frequencies, with a characteristic dip of bone conduction of about 10 to 15 dB at 2 kHz, the **Carhart notch**. The notch is mechanical, not cochlear: it disappears after successful stapes surgery. Tympanometry is type A or As, and the stapedial reflexes are absent.

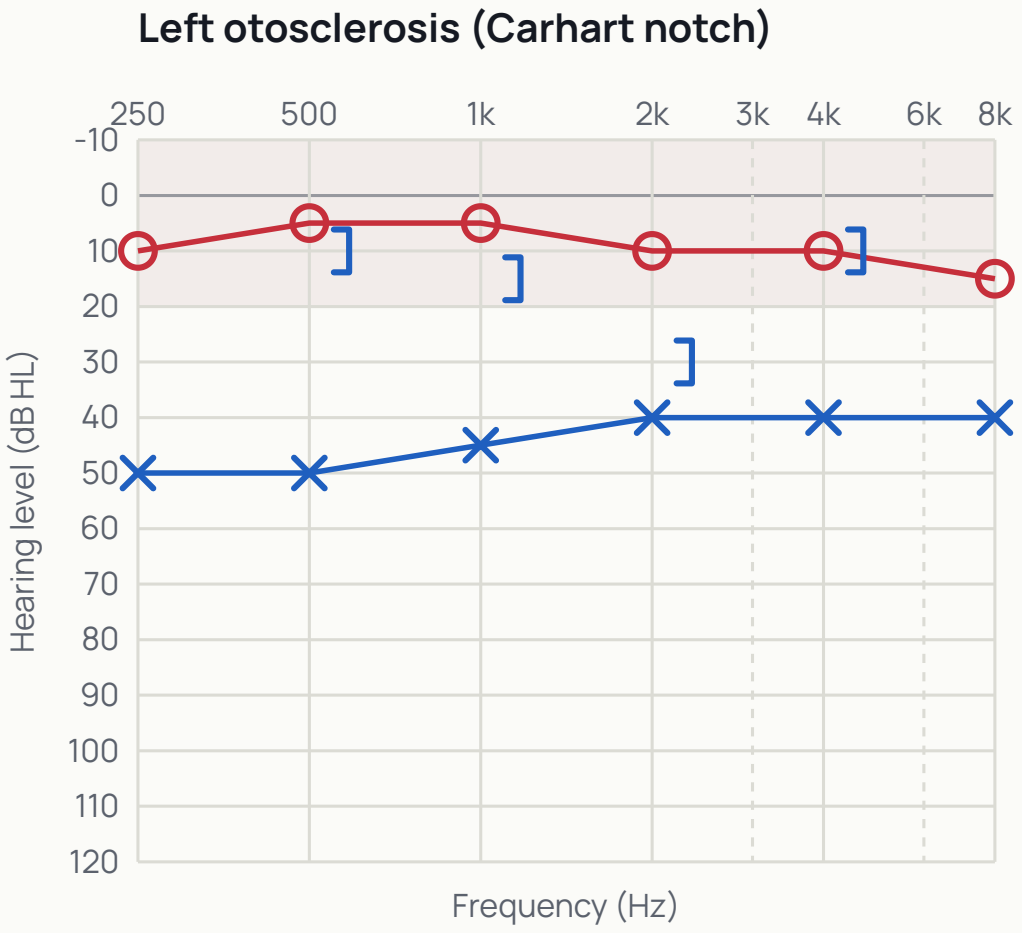


FIGURE 6 Left otosclerosis. Air conduction is about 40 to 50 dB; masked bone conduction is near normal except for the dip at 2 kHz (Carhart notch). Air–bone gap about 35 dB.

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A **mixed loss** combines a raised bone conduction with an air–bone gap, as in chronic otitis media with cochlear damage from long-standing infection, otosclerosis with cochlear involvement, or a temporal bone fracture. Both components matter: the gap may be closed by surgery, but the sensorineural part sets the best possible result, so a patient with bone conduction of 40 dB will still need a hearing aid after a perfect tympanoplasty.

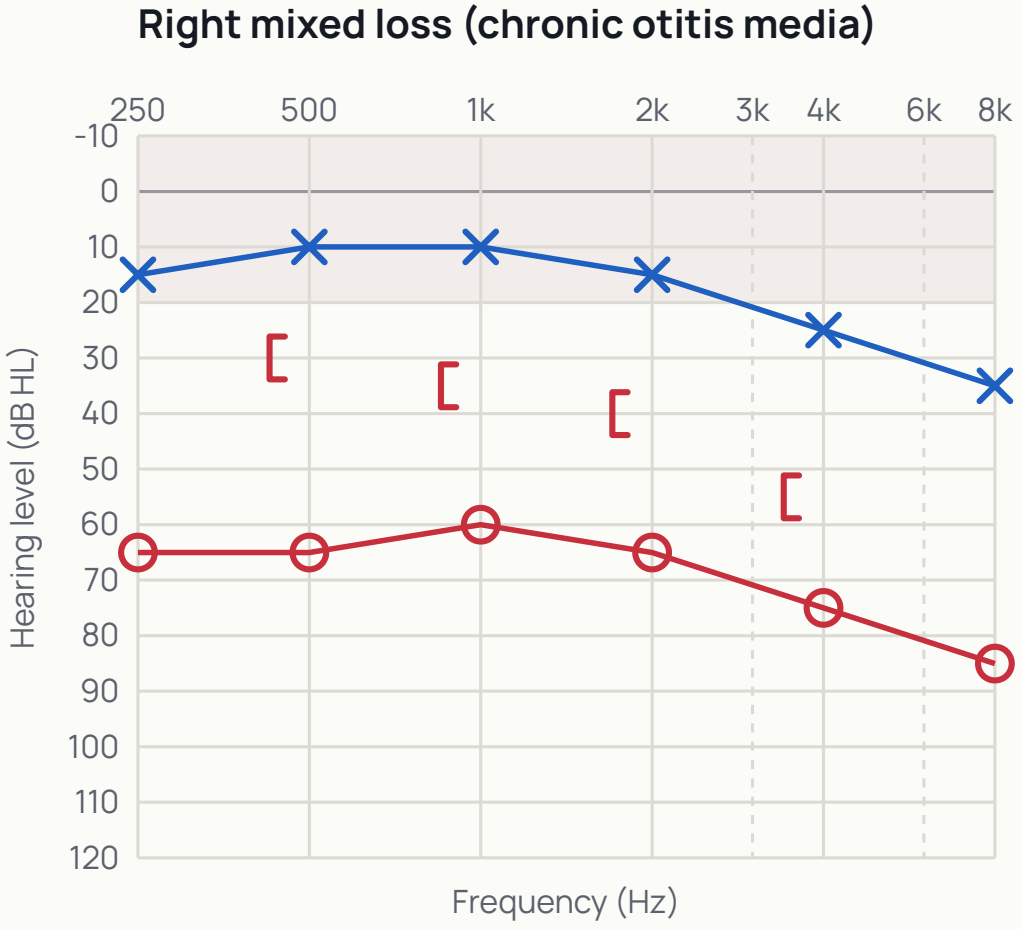


FIGURE 7 Right mixed loss. Masked bone conduction is 30 to 55 dB (the sensorineural part), and air conduction 60 to 85 dB (the gap of about 30 dB is the conductive part).

Drawn for ENT Atlas

5.2 Sensorineural patterns

The *shape* of a sensorineural loss points to its cause.

SENSORINEURAL SHAPES AND WHAT THEY SUGGEST

SHAPE	TYPICAL CAUSE	CLUES
Bilateral, symmetrical, sloping to the high frequencies	Presbycusis, ototoxicity	Age; aminoglycosides, cisplatin
Notch at 3, 4 or 6 kHz with recovery at 8 kHz	Noise	Occupational or recreational noise; usually bilateral
Low-frequency (rising) loss, fluctuating	Ménière's disease, early hydrops	Episodic vertigo, fullness, roaring tinnitus

SHAPE	TYPICAL CAUSE	CLUES
Mid-frequency "cookie-bite" (U-shaped) loss	Genetic, congenital	Family history; stable
Flat, severe, sudden	Sudden sensorineural loss, viral, vascular	Onset within 72 hours
Asymmetric	Vestibular schwannoma until proven otherwise; also Ménière's, sudden loss, noise from one side (shooting)	MRI of the internal auditory meatus
Dead ear	Labyrinthitis, meningitis, fracture, surgery	Shadow curve without masking

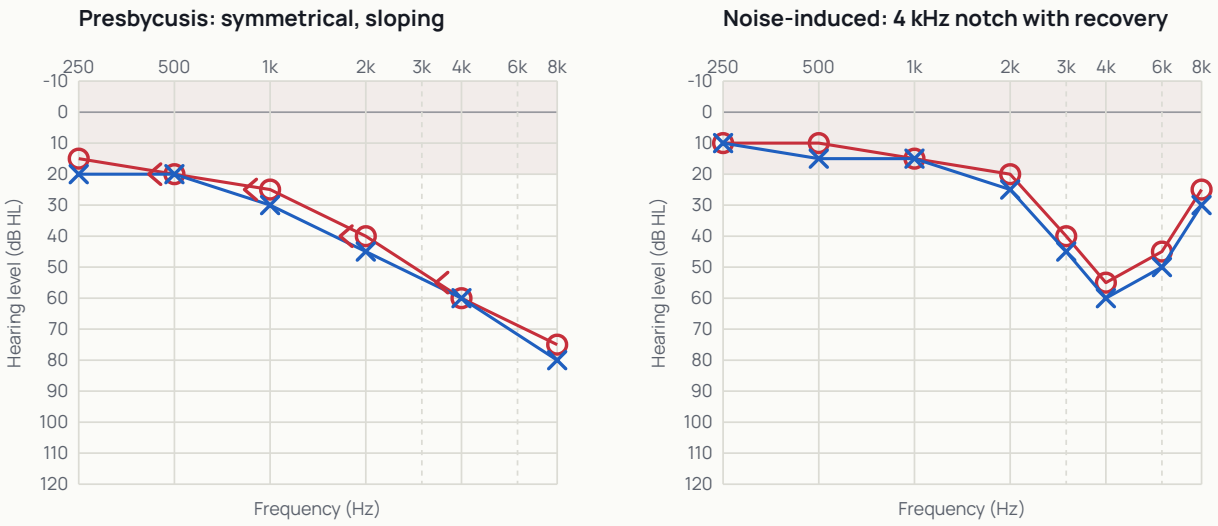
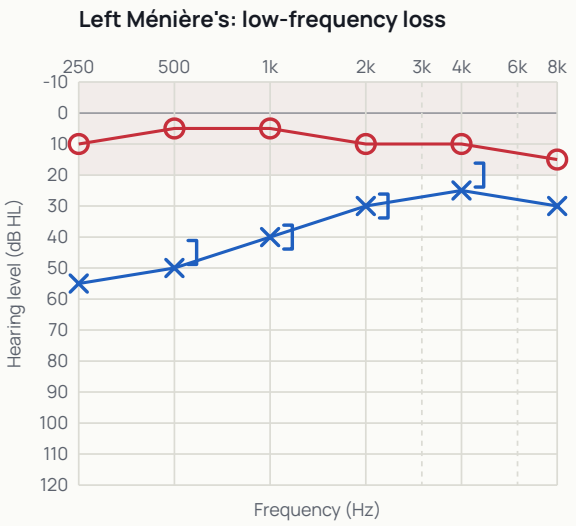


FIGURE 8 Left: presbycusis, a symmetrical high-frequency sloping loss. Right: noise damage, a notch at 4 kHz that recovers at 8 kHz; the recovery separates noise from ageing, although both often coexist.

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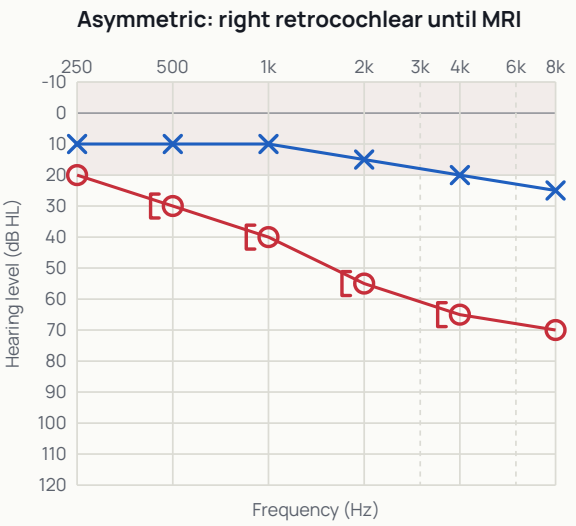


FIGURE 9 Left: early Ménière's disease in the left ear, a low-frequency (rising) sensorineural loss that fluctuates between tests. Right: an asymmetric right sensorineural loss; with no explanation, this needs an MRI.

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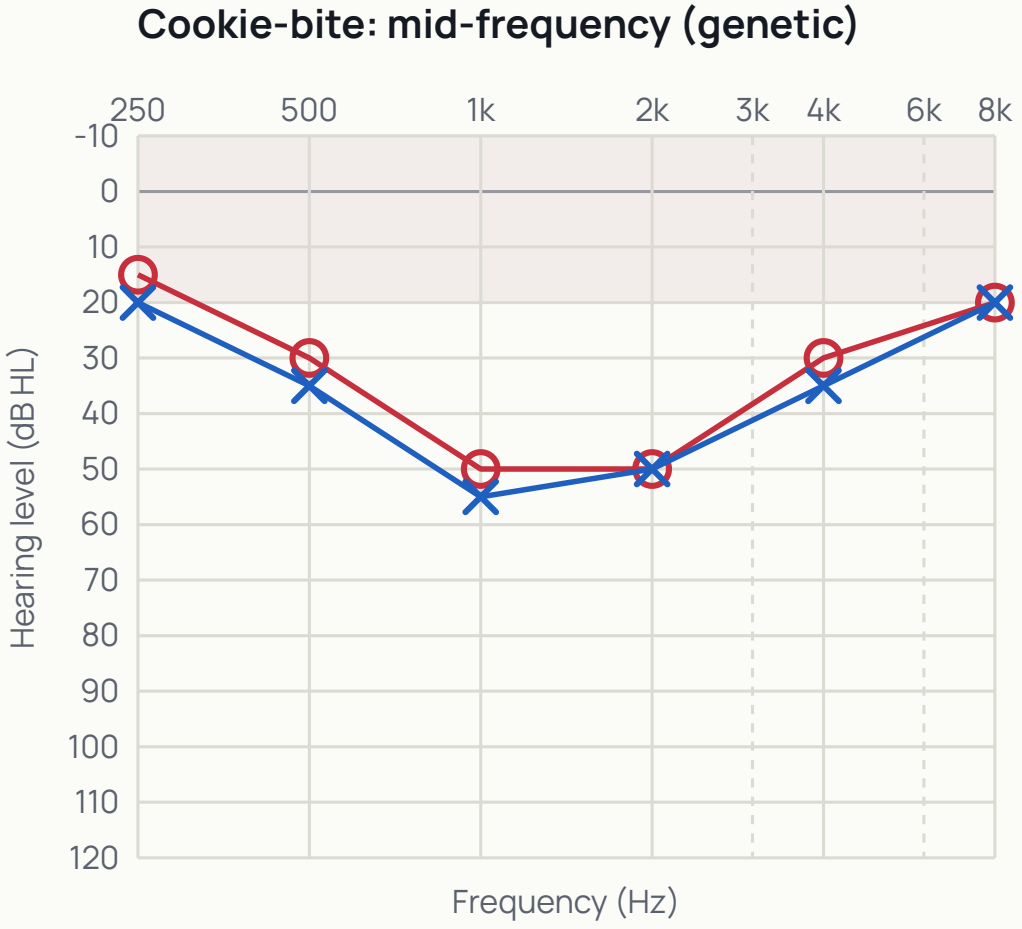


FIGURE 10 A mid-frequency "cookie-bite" loss, typical of some genetic hearing losses. Drawn for ENT Atlas

5.3 When the audiogram does not fit

Non-organic (functional) hearing loss is suggested when thresholds vary between tests, the patient converses easily despite a "severe" loss, the speech reception threshold is much better than the pure-tone average, a unilateral "dead ear" shows no shadow curve, or acoustic reflexes are present at levels near or below the claimed threshold. Objective tests (otoacoustic emissions, ABR or cortical evoked response audiometry) establish the true thresholds; the Stenger test detects a feigned unilateral loss.

6 CHAPTER SIX Speech audiometry

Understanding, not just hearing.

6.1 Speech reception threshold

The level at which half of two-syllable spondee words are repeated correctly; it lies within 10 dB of the pure-tone average.

6.2 Speech discrimination

Single-syllable words presented 30 to 40 dB above the SRT. Good in conductive loss, reduced in cochlear loss, very poor in retrocochlear loss.

6.3 Roll-over

Discrimination falling as loudness increases, a sign of a retrocochlear lesion.

6.4 Reading the speech audiogram

The speech audiogram plots the percentage of words repeated correctly against loudness. In a **conductive loss** the curve has a normal shape but is shifted to the right: once the sound is loud enough, the patient understands everything. In a **cochlear loss** the curve is shifted and may never reach 100%, because the damaged cochlea distorts the sound. In a **retrocochlear loss** the maximum is poor and falls again at high levels (roll-over), and speech discrimination is much worse than the pure-tone thresholds would suggest. Speech scores matter clinically: they predict hearing aid benefit and help select candidates for cochlear implants, which are assessed with sentence tests in quiet and in noise.

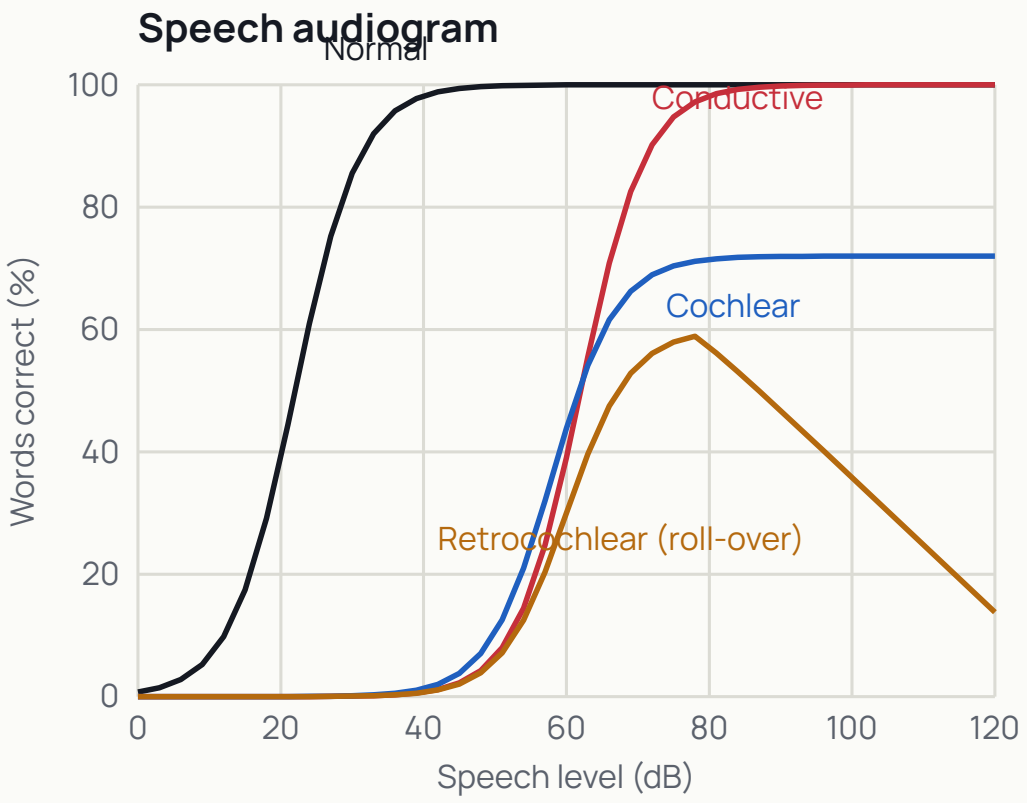


FIGURE 11 Speech audiogram patterns: normal, conductive (shifted but reaching 100%), cochlear (shifted and reduced maximum) and retrocochlear (poor maximum with roll-over).
Drawn for ENT Atlas

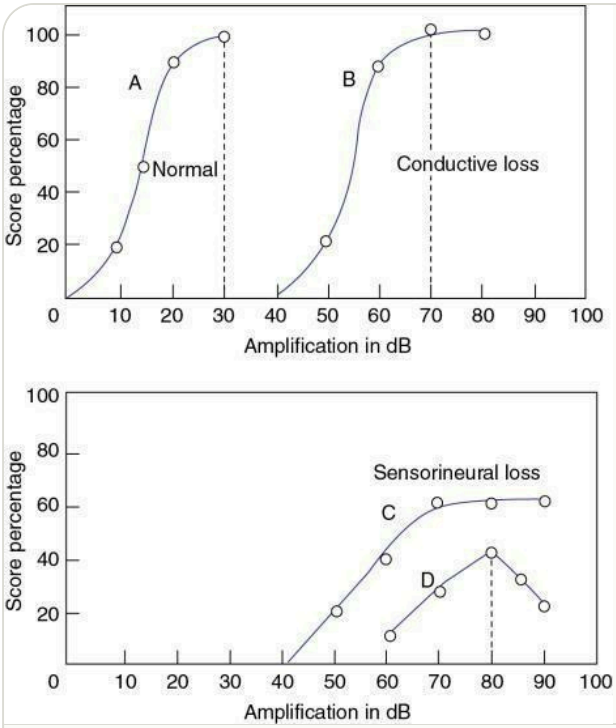


FIGURE 12 Speech audiogram.
Dhingra, fig. 4.3

7 CHAPTER SEVEN Impedance audiometry

The middle ear and reflexes.

7.1 Tympanometry

A probe in the canal changes the pressure while measuring how much sound is reflected from the drum. The peak of compliance shows the middle ear pressure and the mobility of the system.

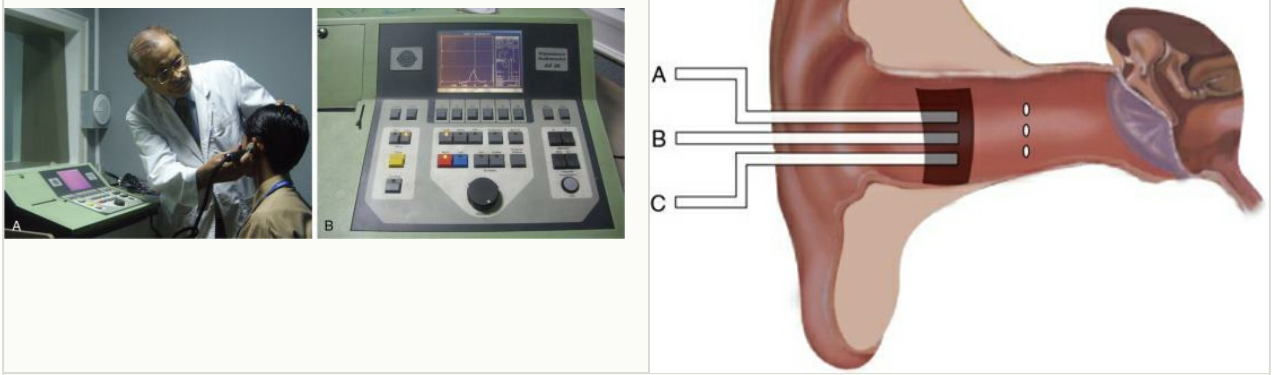


FIGURE 13 First, impedance audiometry in progress. Second, the principle of the impedance audiometer.

Dhingra, figs 4.4 and 4.5

TYMPANOGRAM TYPES

TYPE	FINDING	THINK OF
A	Normal peak	Normal ear
As	Low peak	Otosclerosis, malleus fixation
Ad	High peak	Ossicular discontinuity, lax drum
B	Flat	Fluid, thick drum; perforation or patent grommet if the volume is high
C	Peak below −100 mm H2O	Retraction, tube dysfunction

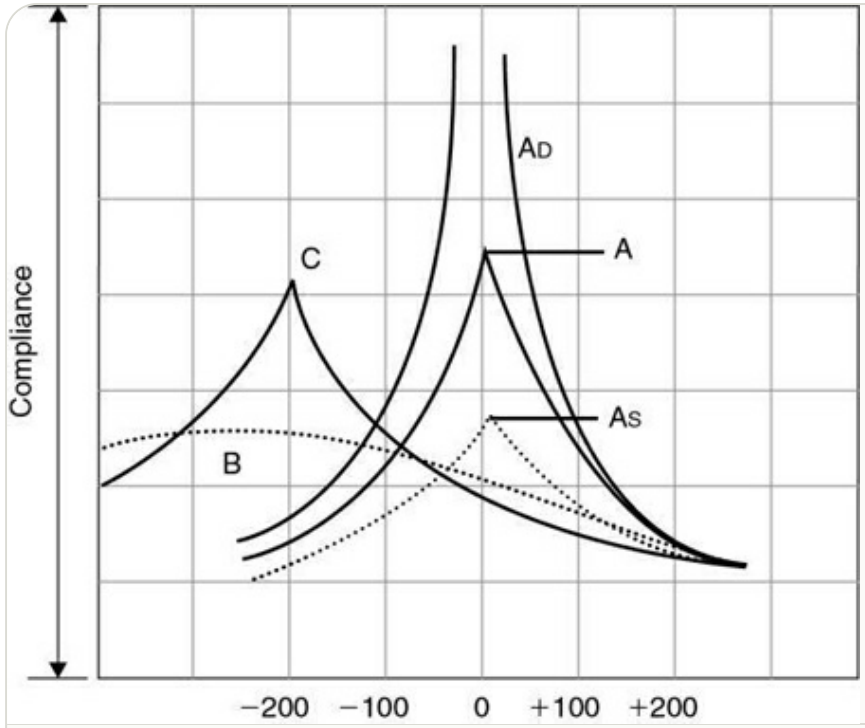


FIGURE 14 Types of tympanogram.
Dhingra, fig. 4.6

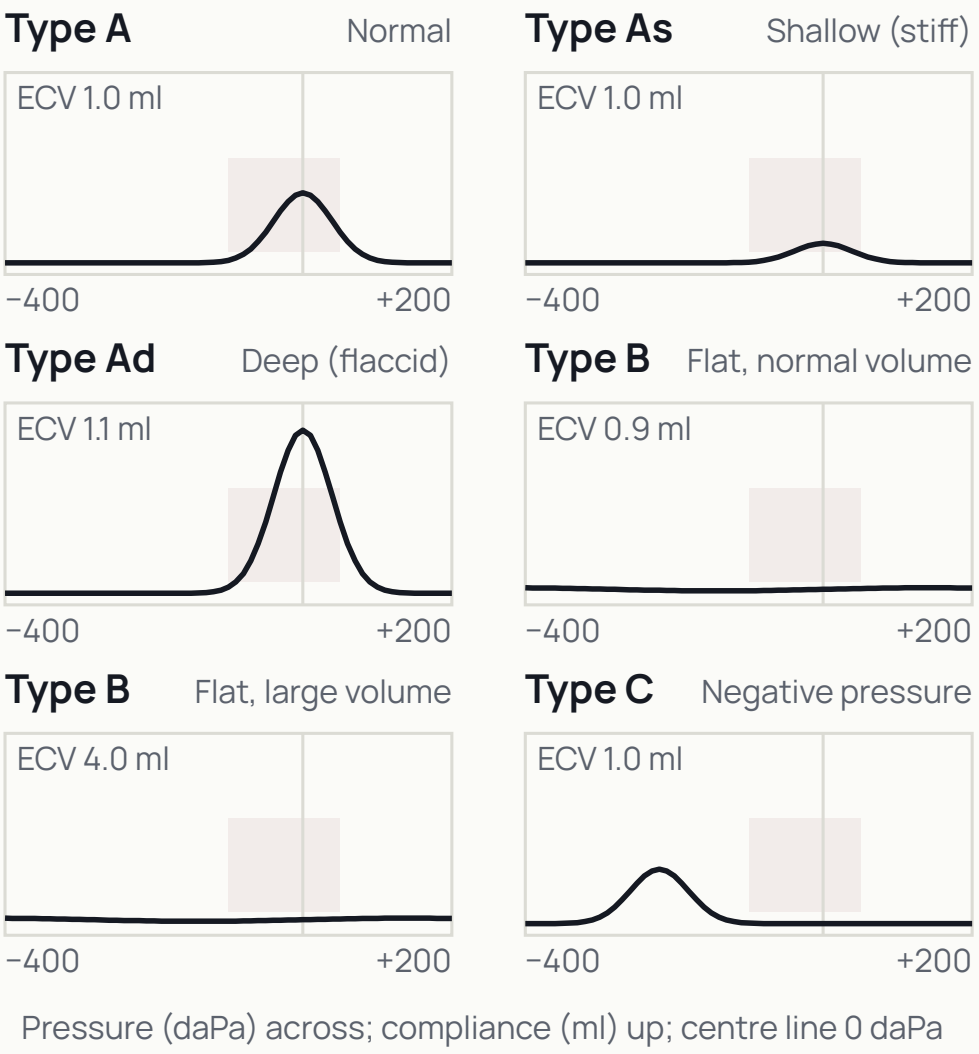


FIGURE 15 Reading a tympanogram: the peak height (compliance), the peak position (middle ear pressure) and the ear canal volume (ECV). The shaded box is the normal range, roughly −100 to +50 daPa and 0.3 to 1.5 ml.

Drawn for ENT Atlas

7.2 Reading a tympanogram

Read three numbers, not just the shape. The **peak pressure** shows the middle ear pressure; normal is about −100 to +50 daPa (1 daPa is about 1 mm H_2O). The **peak compliance** shows how mobile the drum and chain are; normal in adults is about 0.3 to 1.5 ml. The **ear canal volume** is the air between the probe and the drum; normal is about 0.6 to 1.5 ml in adults and 0.3 to 0.9 ml in children.

A FLAT TYMPANOGRAM: WHAT THE VOLUME TELLS YOU

EAR CANAL VOLUME	MEANING
Normal	Middle ear fluid or a thickened, immobile drum

EAR CANAL VOLUME	MEANING
High (over about 2 ml adults, 1 ml children)	Perforation or a patent ventilation tube
Low	Wax or debris against the drum, or the probe against the canal wall

An ear canal volume over about 2.5 mL in adults (2 mL in children) is a strong sign of a perforation or patent grommet. Infants under six months need a 1000 Hz probe tone, because the soft canal wall makes the standard 226 Hz test unreliable. A patulous Eustachian tube produces a trace that moves with breathing; a glomus tumour can produce pulsations in time with the heartbeat.

7.3 Stapedial (acoustic) reflex

A loud sound, 70 to 100 dB above threshold, contracts the stapedius in both ears.

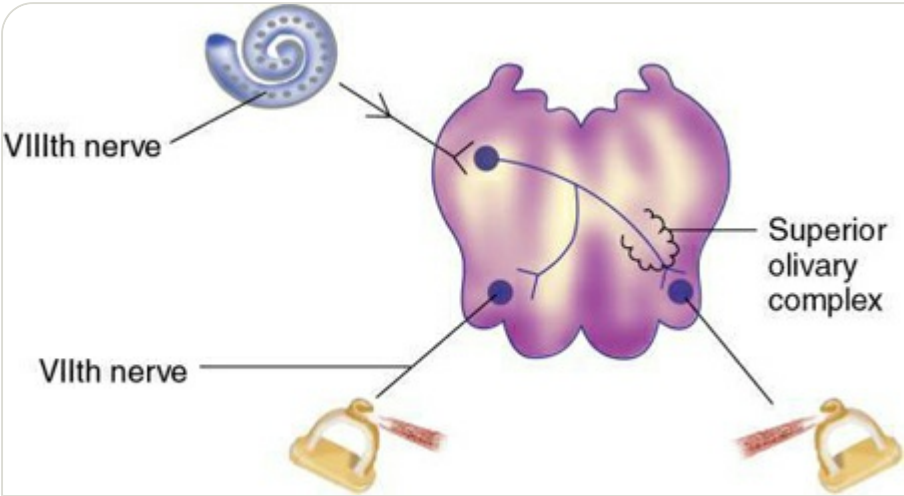


FIGURE 16 The acoustic reflex pathway.
Dhingra, fig. 4.7

7.3.1 What reflexes tell you

Reflexes are measured in four conditions: ipsilateral and contralateral for each ear. Record the **stimulus ear** and the **probe ear** carefully: a contralateral right reflex means sound in the right ear, response measured in the left.

- A reflex at only 40 to 60 dB above threshold: recruitment, a cochlear lesion.
- Reflex decay to half within 10 seconds: an eighth nerve lesion.
- Absent with normal hearing: a facial nerve lesion above the nerve to stapedius.

- Present on the same side but absent on the other: a brainstem lesion.
- Absent: stapes fixation or any middle ear disease.

REFLEX PATTERNS AND THE SITE OF THE LESION

FINDING	SITE
Absent whenever the probe is in the affected ear	Middle ear on that side (the probe cannot detect a reflex)
Absent whenever the affected ear is stimulated	Severe cochlear loss or eighth nerve lesion on that side
Absent whenever the probe is in the affected ear, with normal hearing	Facial nerve on that side, above the nerve to stapedius
Ipsilateral reflexes present, both contralateral reflexes absent	Brainstem lesion crossing the midline
Present with an apparent severe loss	Non-organic loss

CHAPTER EIGHT

8 Site-of-lesion tests

Cochlear or retrocochlear?

Older behavioural tests, now largely replaced by MRI:

- **Alternate binaural loudness balance:** recruitment in cochlear loss.
- **Short increment sensitivity index (SISI):** high scores in cochlear loss.
- **Tone decay:** over 25 dB in retrocochlear loss.
- **Bekesy audiometry.**

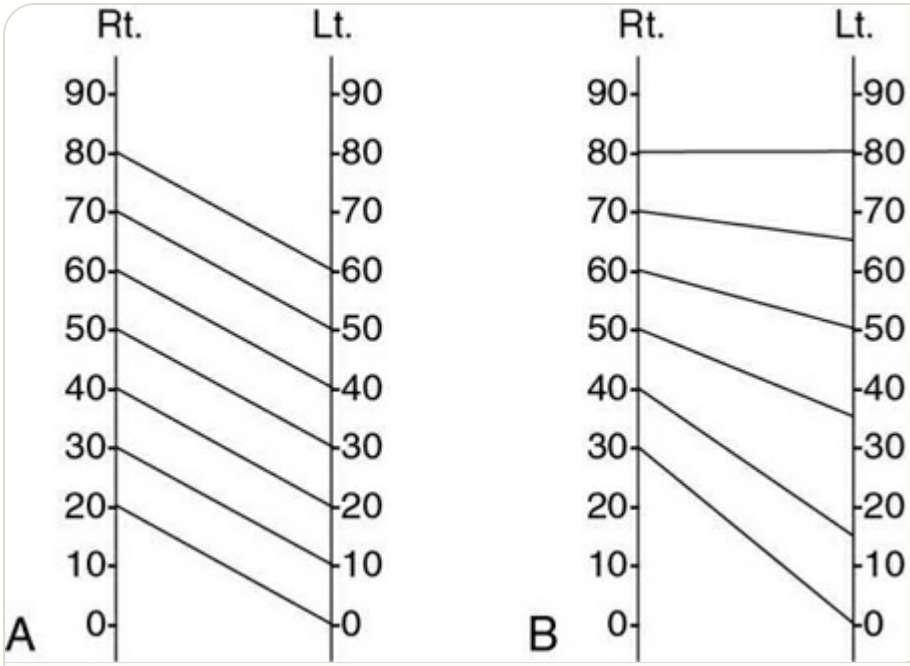


FIGURE 17 Alternate binaural loudness balance test for recruitment.
Dhingra, fig. 4.8

9 CHAPTER NINE Objective tests

When the patient cannot respond.

9.1 Otoacoustic emissions

Sounds produced by the outer hair cells and recorded in the canal. Transient emissions disappear with losses over about 30 dB, distortion-product emissions over about 50 dB. Absent with middle ear fluid or debris; present in auditory neuropathy. Used for newborn screening.

Results are reported as **pass** (clear response) or **refer**. A refer is not a diagnosis of deafness: it may mean vernix or fluid in the canal, a noisy baby or environment, or a true loss, which is why screening repeats the test and then uses automated ABR. Because emissions reflect outer hair cell function, they can show early damage from noise or ototoxic drugs before the audiogram changes, and they monitor patients on cisplatin.

9.2 Auditory brainstem response

Five to seven waves recorded from the scalp within 10 ms of a click: I and II from the eighth nerve, III the cochlear nucleus, IV the superior olive, V the lateral lemniscus and inferior colliculus (*E. COLI*). Used for thresholds in infants (pass at about 30 to 40 dB nHL) and to look for retrocochlear lesions, though it misses small tumours. Unaffected by sleep and sedation.

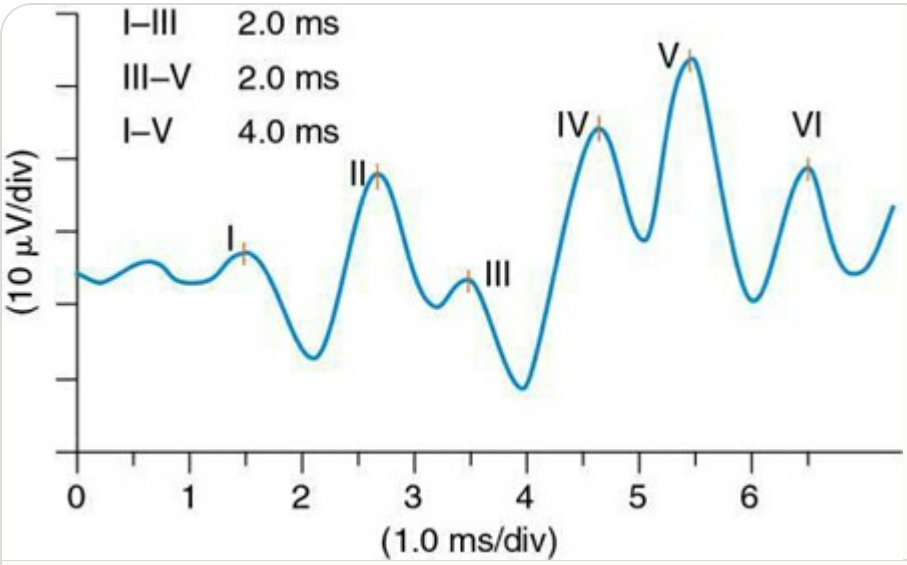


FIGURE 18 Brainstem auditory evoked potentials.
Dhingra, fig. 4.10

9.3 Reading an ABR

Look at three things. **Wave latencies:** wave V normally appears at about 5.5 to 6 ms with a loud click. **Interpeak intervals:** I–III about 2 ms, III–V about 2 ms, and I–V about 4 ms; the I–V interval measures conduction along the nerve and brainstem and is *independent of the middle ear*. **Threshold:** the softest level at which wave V can still be identified, used to estimate hearing in infants. In a conductive loss all the waves are delayed but the I–V interval is normal; in a cochlear loss wave I may be small or absent and the latency of V falls steeply as loudness rises (recruitment); in a retrocochlear lesion the I–V interval is prolonged or the waves after I are lost, and wave V on the affected side is delayed compared with the other ear.

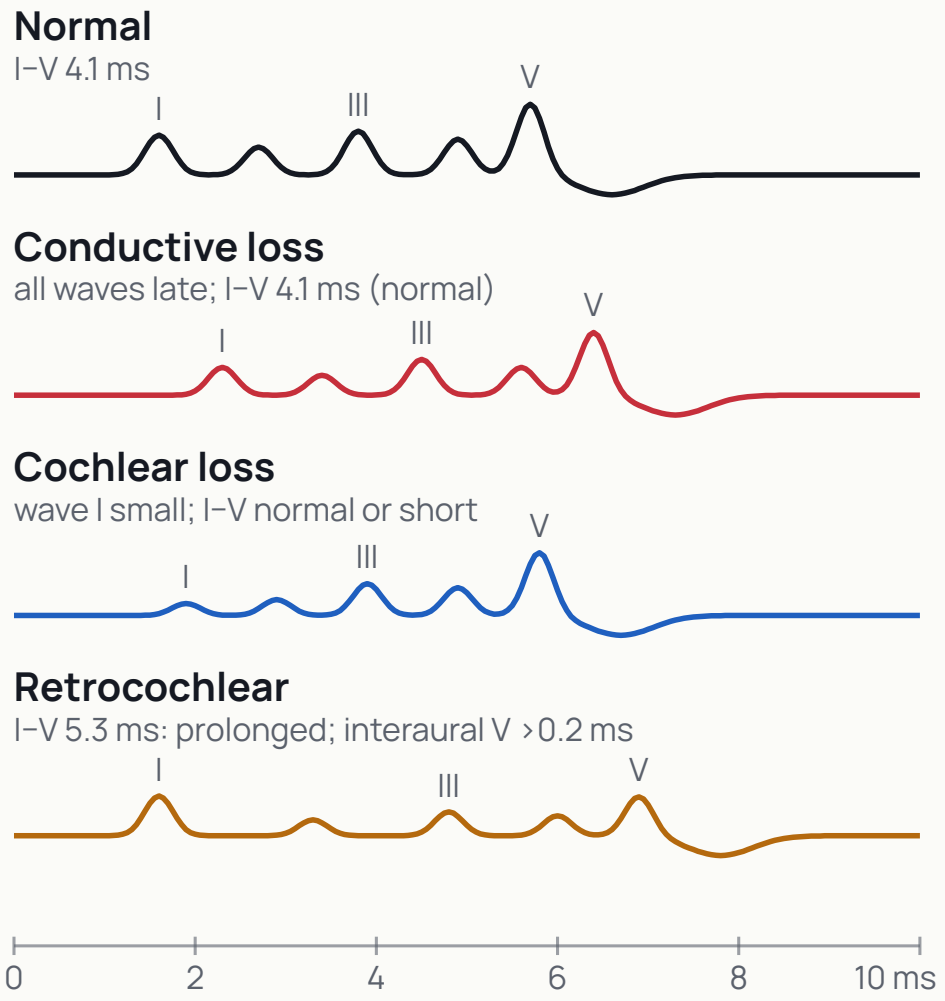


FIGURE 19 ABR traces. Conductive loss delays every wave; a retrocochlear lesion prolongs the I–V interval. Values are typical, not normative.

Drawn for ENT Atlas

WHERE THE BOOKS DIFFER · ABNORMAL INTERAURAL WAVE V DIFFERENCE

Dhingra	Lee
Over 0.2 ms.	0.4 to 0.5 ms.

9.4 Auditory steady-state response

Frequency-specific thresholds into the profound range, beyond the reach of ABR (about 80 dB).

9.5 Electrocochleography

Records the summating potential, action potential and cochlear microphonic. An enlarged SP to AP ratio suggests Ménière's disease (see that topic).

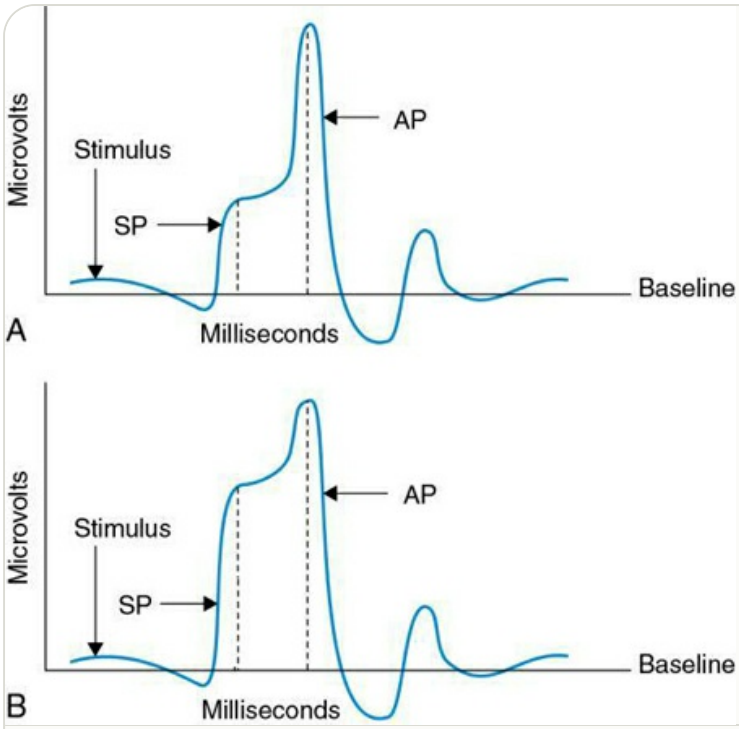
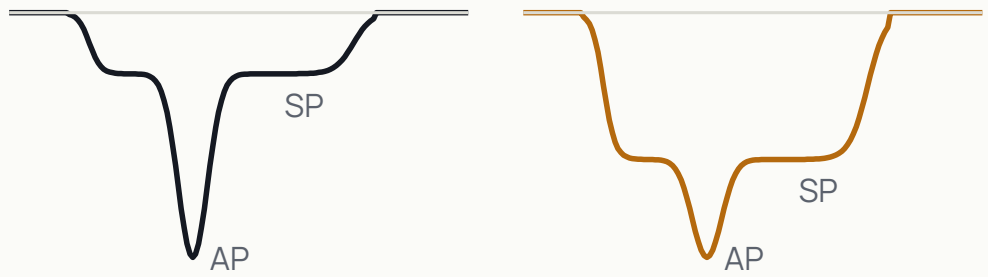


FIGURE 20 Electrocochleography: normal ear and an ear with Ménière's disease.
Dhingra, fig. 4.9



Negative down: SP is the plateau, AP the sharp trough.

FIGURE 21 Reading electrocochleography: the summating potential (SP) is the shoulder before the sharp action potential (AP). An SP/AP ratio above about 0.4 to 0.5 suggests endolymphatic hydrops; a normal result does not exclude it.

Drawn for ENT Atlas

9.6 Auditory neuropathy

Present otoacoustic emissions and cochlear microphonic with an absent or abnormal ABR: the hair cells work but the nerve does not carry the signal synchronously.

10 CHAPTER TEN Putting the tests together

Patterns in common conditions.

THE TEST BATTERY IN COMMON CONDITIONS

CONDITION	TUNING FORKS	AUDIOGRAM	TYMPANOGRAM	REFLEXES	OTHER
Wax or otitis media with effusion	Rinne negative, Weber to the ear	Conductive, flat or low-frequency	B, normal volume (effusion)	Absent	OAE absent

CONDITION	TUNING FORKS	AUDIOGRAM	TYMPANOGRAM	REFLEXES	OTHER
Perforation	Rinne negative if gap over about 20 dB	Conductive	B, high volume	Absent	—
Otosclerosis	Rinne negative	Conductive, Carhart notch	A or As	Absent	Bone better than normal at 2 kHz after surgery
Ossicular discontinuity	Rinne negative	Conductive, up to 60 dB	Ad	Absent	—
Superior canal dehiscence	Weber to the ear	Low-frequency gap, bone better than normal	A	Present	VEMP thresholds low; CT
Presbycusis	Rinne positive	Bilateral high-frequency sensorineural	A	Present, may be at reduced levels	Speech scores fair
Ménière's disease	Weber away	Low-frequency sensorineural, fluctuating	A	Present, recruitment	ECochG SP/AP raised
Vestibular schwannoma	Weber away	Asymmetric sensorineural	A	Absent or decay	Poor speech, roll-over; ABR I–V prolonged; MRI
Auditory neuropathy	Variable	Any, often fluctuating	A	Absent	OAE present, ABR absent or abnormal
Non-organic loss	Inconsistent	Variable, no shadow curve	A	Present near claimed threshold	SRT better than PTA; objective tests normal

11 CHAPTER ELEVEN Testing children

By age.

- Birth to six months: behavioural observation, OAE and ABR.
- Six to thirty months: visual reinforcement audiometry.
- From about thirty months: play audiometry.
- Older children: conventional audiometry. Normal thresholds in children are 15 dB or better.

Visual reinforcement audiometry conditions the child to turn towards a sound, rewarded by a light or puppet, and uses insert earphones or free field; free-field results reflect the better ear only. **Play audiometry** turns the test into a game, such as putting a peg in a board each time a sound is heard. Tympanometry and otoacoustic emissions support behavioural tests at every age, and ABR under sleep or sedation is used when behavioural results are unreliable.

12 CHAPTER TWELVE For the examination

What the examiner wants to hear.

12.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Rinne and Weber:** technique and interpretation, and the false-negative Rinne.
- **Masking:** when and why; the plateau method, shadow curve and masking dilemma.
- **Reading an audiogram aloud:** describe each ear, the type, degree and shape of loss, and what you would do next.
- **Audiogram patterns:** Carhart notch, noise notch, low-frequency loss, asymmetry, dead ear.
- **Tympanogram types** and the ear canal volume.
- **Reflex patterns** and the site of the lesion.
- **Reading an ABR:** latencies, the I–V interval and wave V asymmetry.
- **Uses of the stapedial reflex.**

- **OAE versus ABR,** and the waves of the ABR.
- **Auditory neuropathy.**

12.2 Common pitfalls

COMMON PITFALLS

- Reading a false-negative Rinne as a conductive loss.
- Forgetting to mask.
- Calling a flat tympanogram fluid without checking the canal volume.
- Taking a shadow curve for a real moderate loss in a dead ear.
- Planning stapes surgery for an air–bone gap with present reflexes before excluding a third window.
- Accepting a severe loss in a patient who follows quiet conversation.

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